DAELIM

SAFETY DATA SHEET

Product Name : MAA

Emergency Telephone Number
+82-61-688-6310/6375(24hours)

Established Date : 2015.12.17

Revision No. : 1 - 03 (Revision Date : 2024.01.10

1. IDENTIFICATION

A. PRODUCT NAME
- Methacrylic Acid

B. Recommended Use and Restriction on Use
- General use : Not available
- Restriction on Use : Not available

C. Information of Manufacturer
- Company Name : DAELIM Industrial Co., Ltd.
- Address : 2, Yeosusandan 3-ro, Yeosu-si, Jeollanam-do, 59612, Korea
- Person in charge : PB Production Team 1, Person in charge of (M)SDS
- Telephone number : +82-61-688-6310/6375
- Fax number : +82-61-688-6379

2. HAZARD IDENTIFICATION

A. GHS Classification :
- Flammable liquids (Category 4), H227
- Acute toxicity, Oral (Category 4), H302
- Acute toxicity, Inhalation (Category 4), H332
- Acute toxicity, Dermal (Category 3), H311
- Skin corrosion (Category 1), H314

B. GHS label elements
- Hazard symbols :



- Signal word : Signal word: DANGER
- Hazard statement :
    - H227 Combustible liquid.
    - H302 + H332 Harmful if swallowed or if inhaled
    - H311 Toxic in contact with skin.
    - H314 Causes severe skin burns and eye damage.
- Precautionary statements :
- Hazards not otherwise classified (HNOC) None identified
    - Prevention
        - Wash face, hands and any exposed skin thoroughly after handling
        - Do not eat, drink or smoke when using this product
        - Wear protective gloves/protective clothing/eye protection/face protection
        - Use only outdoors or in a well-ventilated area
        - Do not breathe dust/fume/gas/mist/vapors/spray
        - Keep away from heat/sparks/open flames/hot surfaces. - No smoking
        - Keep cool
    - Response
        - Immediately call a POISON CENTER or doctor/physician
    - Inhalation
        - IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing
    - Skin
        - Wash contaminated clothing before reuse
        - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower
    - Eyes
        - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continuerinsing
    - Ingestion
        - Rinse mouth
        - Do NOT induce vomiting
    - Fire
        - In case of fire: Use CO2, dry chemical, or foam for extinction
    - Storage
        - Store locked up
        - Store in a well-ventilated place. Keep container tightly closed
    - Disposal
        - Dispose of contents/container to an approved waste disposal
- Hazards not otherwise classified (HNOC): None identified

3. COMPOSION/INFORMATION ON INGREDIENTS

Chemical Name	Generic name	CAS No. or Reference No.	Contents (%)
Methacrylic Acid	MMA	79-41-4	99.5
Monomethyl ether of hydroquinone	MEHQ	150-76-5	0.5

4. FIRST-AID MESURES

A. Eye Contact

- Rinse thoroughly with plenty of water for at least 15 minutes and consult a physician.
- Continue rinsing eyes during transport to hospital.

B. Skin Contact

- Take off contaminated clothing and shoes immediately.
- Wash off with soap and plenty of water.
- Take victim immediately to hospital. Consult a physician.

C. Inhalation

- If not breathing, give artificial respiration.
- Do not use mouth-to-mouth method if victimingested or inhaled the substance; give artificial respiration with the aid of a pocket maskequipped with a one-way valve or other proper respiratory medical device.
- Removetofreshair.
- Immediate medical attention is required.

D. Ingestion

- Do not induce vomiting.
- Never give anything by mouth to an unconscious person.
- Rinse mouth with water.
- Consult a physician

E. Notes to Physician

- Treat symptomatically.

5. FIRE FIGHTING MEASURES

A. Suitable (Unsuitable) extinguishing media

- CO_2, dry chemical, dry sand, alcohol-resistant foam. Water mist may be usedtocool closed

containers.

B. Specific hazards arising from the chemical

- Thermal decomposition can lead to release of irritating gases and vapors.
- The product causes burns of eyes, skin andmucousmembranes.
- Combustible material.
- Containers may explode when heated.

C. Hazardous Combustion Products

- Carbon monoxide (CO). Carbon dioxide (CO2).

D. Advice for firefighters: Hazardous Combustion Products

- Wear self-contained breathing apparatus for firefighting if necessary

6. ACCIDENTAL RELEASE MEASURES

A. Personal Precautions, Protective Equipment and Emergency procedures

- Use personal protective equipment as required.
- Ensure adequate ventilation.
- Evacuatepersonnel to safe areas.
- Keep people away from and upwind of spill/leak.
- Remove all sources of ignition.
- Take precautionary measures against static discharges

B. Environmental Precautions

- Prevent further leakage or spillage if safe to do so.
- Do not let product enter drains.
- Discharge into the environment must be avoided.

C. Products Methods and materials for containment and cleaning up

- Contain spillage, and then collect with an electrically protected vacuum cleaner or by wet-brushing and place in container for disposal according to local regulations
- Keep in suitable, closed containers for disposal.

7. HANDLING AND STORAGE

A. Handling

- Wear personal protective equipment/face protection.
- Do not get in eyes, on skin, or onclothing.
- Use only under a chemical fume hood.
- Do not breathe mist/vapors/spray. Donotingest. If swallowed then seek immediate medical assistance.

- Keep away fromopenflames, hot surfaces and sources of ignition.

B. Storage Precautionary Statements :

- Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosivesarea.
- Store indoors.
- Keep away from heat, sparks and flame. Keep at temperatures between18°and 30 °C. Incompatible Materials.
- Strong oxidizing agents. Alkali metals. Strongbases.Finely powdered metals. Ammonia.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

A. Engineering Controls

- A system Investigate engineering techniques to reduce exposures below airborne exposure limits. Provide ventilation if necessary to control exposure levels below airborne exposure limits (see below). If practical, use local mechanical exhaust ventilation at sources of air contamination such as open process equipment. Consult ACGIH ventilation manual or NFPA Standard 91 for design of exhaust systems.

B. Eye/Face Protection

- Tightly fitting safety goggles. Faceshield (8-inch minimum). Use equipment for eye protection tested and approved under appropriate government standards such as NIOSH (US) or EN 166(EU).

C. Skin protection:

- Handle with gloves. Gloves must be inspected prior to use. Use proper glove removal technique (without touching glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices. Wash and dry hands

D. Respiratory protection:

- Where risk assessment shows air-purifying respirators are appropriate use a full-face respirator with multipurpose combination (US) or type ABEK (EN 14387) respirator cartridges as a backup to engineering controls. If the respirator is the sole means of protection, use a full-face supplied air respirator. Use respirators and components tested and approved under appropriate government standards such as NIOSH (US) or CEN (EU).

9. PHYSICAL AND CHEMICAL PROPERTIES

A. Appearance : clear, liquid

B. Odor : Not available

C. Odor threshold : Not available

D. pH : 2.0 - 2.2 at 100 g/l at 20°C (68°F)
E. Melting point/Freezing point : Melting point/range: 12 - 16°C (54 - 61°F) - lit.
F. Initial Boiling Point/Boiling Ranges 163°C (325°F) - lit.
G. Flash point : 77°C (171°F) - closed cup
H. Evapourating Rate : Not available
I. Flammability(solid, gas) : Not available
J. Upper explosive limits: 8.7 %(V)
K. Lower explosive limits : 1.6 %(V)
L. Vapour pressure : 1 hPa (1 mmHg) at 20°C (68°F)
M. Solubility : Soluble
N. Vapour density : 2.97 - (Air = 1.0)
O. Relative density : 1.015g/cm3at25°C(77°F)
P. Partition coefficient of n-octanol/water : log Pow: 0.93
Q. Autoignition Temperature : Not available
R. Decomposition Temperature : Not available
S. Viscosity : 1.4 mPa s at 20 °C
T. Molecular weight 86.09

10. STABILITY AND REACTIVITY

A. Reactivity: Not available

B. Chemical stability
- Stable under recommended storage conditions
- Contains the following stabilizer(s): Mequinol (≥ 200 - ≤ 300 ppm)

C. Conditions to Avoid
- Heat, flames and sparks

D. Incompatible materials
- This material polymerizes exothermically in the presence of heat, contamination, oxygen free atmosphere, free radicals, peroxides, and inhibitor depletion liberating heat

E. Hazardous Decomposition Products
- Oxides of carbon can be liberated at temperatures above ambient.

11. TOXICOLOGICAL INFORMATION

A. Information on toxicological effects
- Acute toxicity
  - LD50 Oral - Rat - male - 1,320 mg/kg
  - (OECD Test Guideline 401)

- LC50 Inhalation - Rat - 4 h - 0.9 - 4.7 mg/l
- (OECD Test Guideline 403)
- LD50 Dermal - Rabbit - 500 - 1,000 mg/kg

- Skin corrosion/irritation:
  - Skin - Rabbit
  - Result: Causes severe burns.(OECD Test Guideline 404)

B. Germ cell mutagenicity: Not available

C. Carcinogenicity

- IARC: No component of this product present at levels greater than or equal to 0.1% is identified as probable, possible or confirmed human carcinogen by IARC.
- ACGIH: No component of this product present at levels greater than or equal to 0.1% is identified as a carcinogen or potential carcinogen by ACGIH.
- NTP: No component of this product present at levels greater than or equal to 0.1% is identified as a known or anticipated carcinogen by NTP.
- OSHA: No component of this product present at levels greater than or equal to 0.1% is identified as a carcinogen or potential carcinogen by OSHA.

D. Reproductive toxicity: Not available

12. ECOLOGICAL INFORMATION

A. Ecotoxicological Information:

- Data on this material and/or its components are summarized below.
- Toxicity to fish LC50 - Oncorhynchus mykiss (rainbow trout) - 85 mg/l - 96 h

B. Persistence and degradability :

- .Biodegradability aerobic - Exposure time 28 d
- Result: 86 % - Readily biodegradable

C. Bioaccumulative potential : Not available

D. Mobility in soil Not available

E. Other adverse effects : Not available

- .An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.
- Harmful to aquatic life.

13. DISPOSAL CONSIDERATIONS

Waste Disposal

Incineration is the recommended method for disposal observing all local, state and federal regulations. Note: Chemical additions to, processing of, or otherwise altering this material may make this waste management information incomplete, inaccurate, or otherwise inappropriate. Furthermore, state and local waste disposal requirements may be more restrictive or otherwise different from federal laws and regulations. The user of this product must properly characterize the waste/container generated from the use of this product in accordance with all applicable federal, state and/or local laws and regulations in order to determine the proper disposal of the waste in accordance with all applicable federal, state and/or local laws and regulations.

14. TRANSPORT INFORMATION

A. DOT Name: Methacrylic acid, stabilized

B. DOT Hazard class : 8

C. DOT Packing group : II

D. UN number: UN2531

15. REGULATORY INFORMATION

A. SARA 302 Components: No chemicals in this material are subject to the reporting requirements of SARA Title III, Section 302.

B. SARA 313 Components: This material does not contain any chemical components with known CAS numbers that exceed the threshold (De Minimis) reporting levels established by SARA Title III, Section 313.

C. SARA 311/312 Hazards: Fire Hazard, Acute Health Hazard, Chronic Health Hazard

D. Massachusetts Right to Know:

- This product does contain the following chemical(s), as indicated below, currently on the Massachusetts Right to Know Substance List.
- Methacrylic Acid
- Monomethyl ether of hydroquinone (MEHQ)

E. New Jersey Right to Know

- This product does contain the following chemical(s), as indicated below, currently on the New Jersey Right to Know Substance List.
- Methacrylic Acid
- Monomethyl ether of hydroquinone (MEHQ)

F. Pennsylvania Right to Know

- This product does contain the following chemical(s), as indicated below, currently on the Pennsylvania Hazardous Substance List.

- Methacrylic Acid
- Monomethyl ether of hydroquinone (MEHQ)

16. OTHER INFORMATION

A. Reference

- This SDS is prepared based on the MSDS provided by DAELIM Industrial Co., Ltd., Korea Test & Research Institute, and many other databases and translated into English in accordance with ISHL Article 41 and MOL Public notice 2013-37. This information only concerns the above-mentioned product and does not need to be valid if used with other(s) or in any process. The information is to our best present knowledge correct and complete and is given in good faith but without warranty. It remains the user's own responsibility to make sure that the information is appropriate and sufficiently complete for their particular use of this product.